

Drone Mapping & GIS Analysis Portfolio

Prepared by: Igor Hajducki | DroneCube Analytics

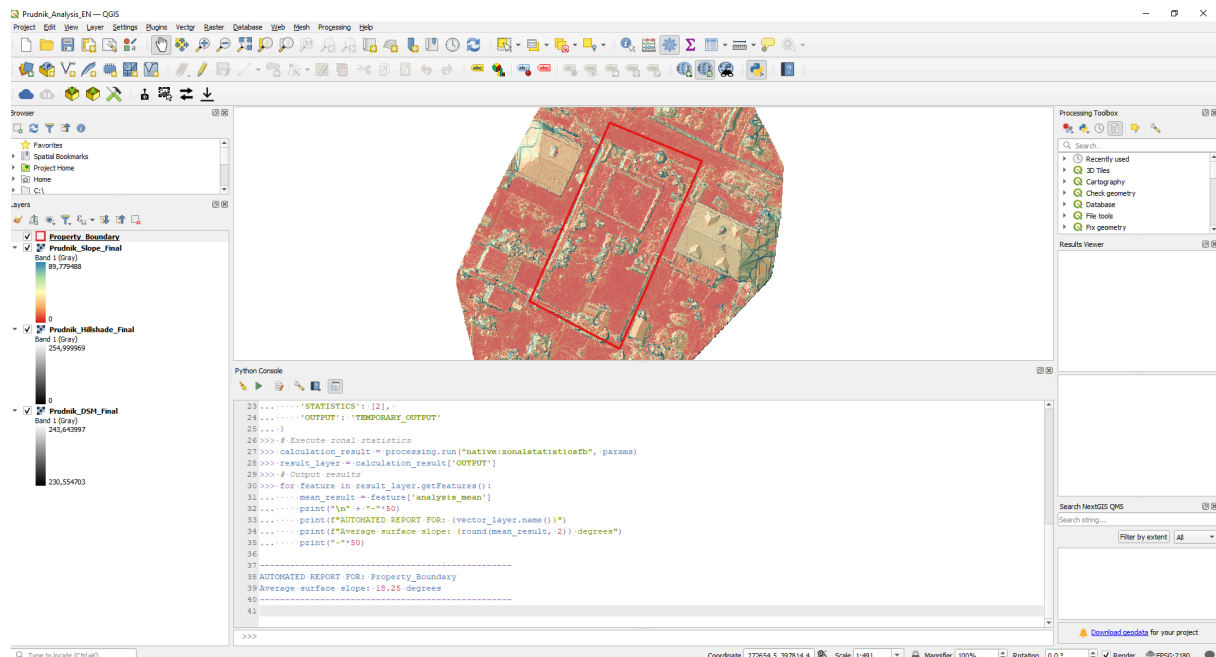
Core Tools: WebODM, QGIS, PyQGIS

Project 1: Automated Zonal Statistics (Python / PyQGIS)

Objective: Eliminate manual error when calculating average terrain/roof slopes for construction and solar potential.

Method: Developed a custom Python script that runs `native:zonalstatisticsfb` in temporary memory to extract exact mathematical averages from raster pixels bounded by a specific vector geometry.

Result: Verified average slope of 18.25°.



```
36
37
38 AUTOMATED REPORT FOR: Property_Boundary
39 Average surface slope: 18.25 degrees
40
41
```

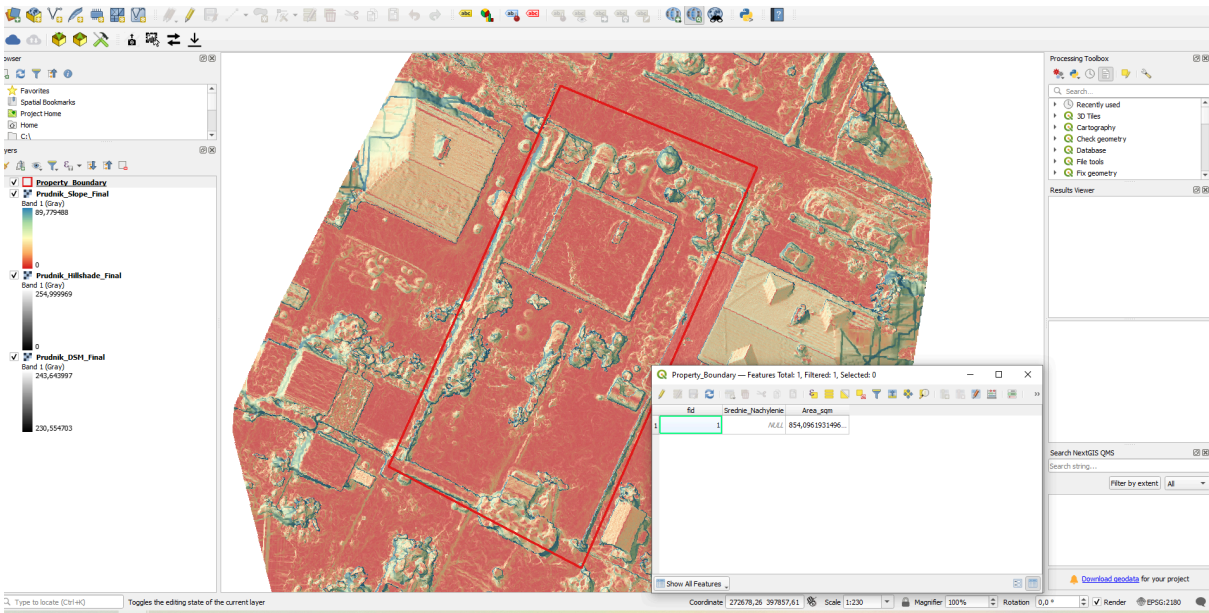
Coordinate 272630,19 397940,88 Scale 1:306 Magnifier 100% Rotation 0,0° Render EPSG:2180

Project 2: Precise Vector Extraction for Progress Tracking

Objective: Replace "visual guessing" with data-driven completion estimates for lender-facing construction reports.

Method: Digitized exact building footprints from high-resolution drone orthomosaics. Utilized QGIS geometry calculators to extract the precise area (\$m^2\$) of current construction phases (e.g., poured foundations) to compare against architectural blueprints.

Result: Verified structure footprint area: 854.10 m².



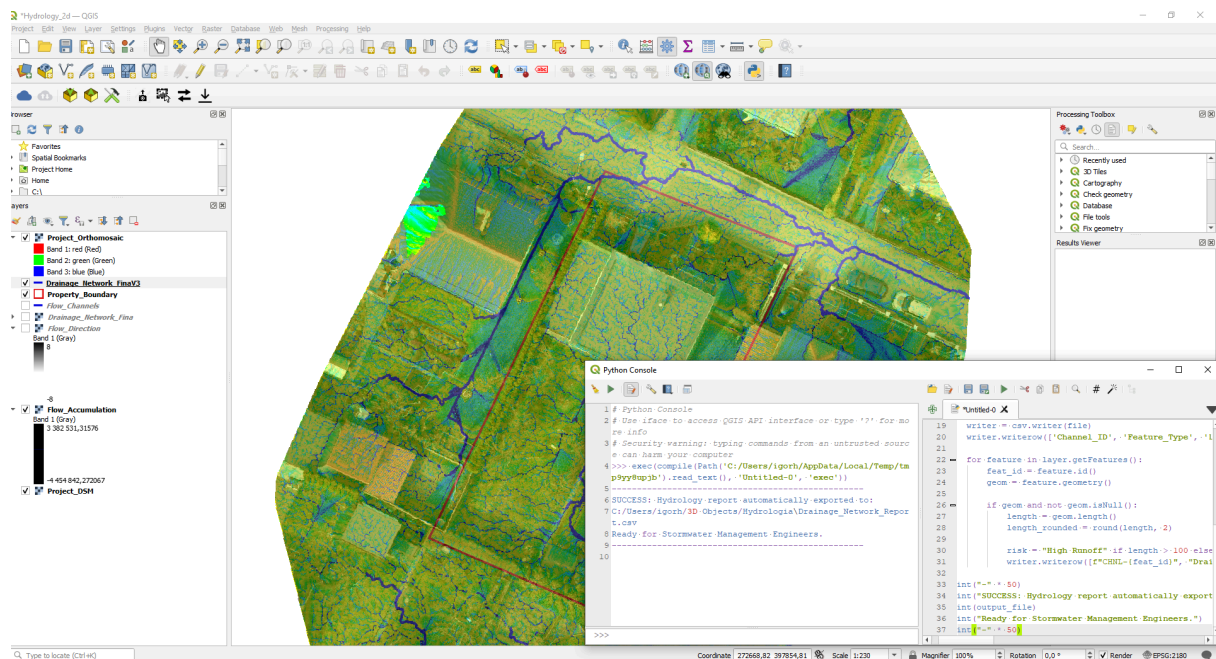
Property_Boundary — Features Total: 1, Filtered: 1, Selected: 0			
	fid	Srednie_Nachylenie	Area_sqm
1	1	NULL	854,0961931496...

Project 3: Automated Hydrological Extraction (Python / PyQGIS)

Objective: Provide engineering-grade stormwater management data by automating the extraction of drainage channel lengths from high-resolution DEMs.

Method: Developed a custom Python script that analyzes flow accumulation vector geometries. The script calculates precise lengths (in meters) for each drainage line and automatically exports a categorized runoff risk report directly to a CSV file for civil engineers.

Result: Verified extraction of drainage network geometries and automated generation of structured CSV reports.



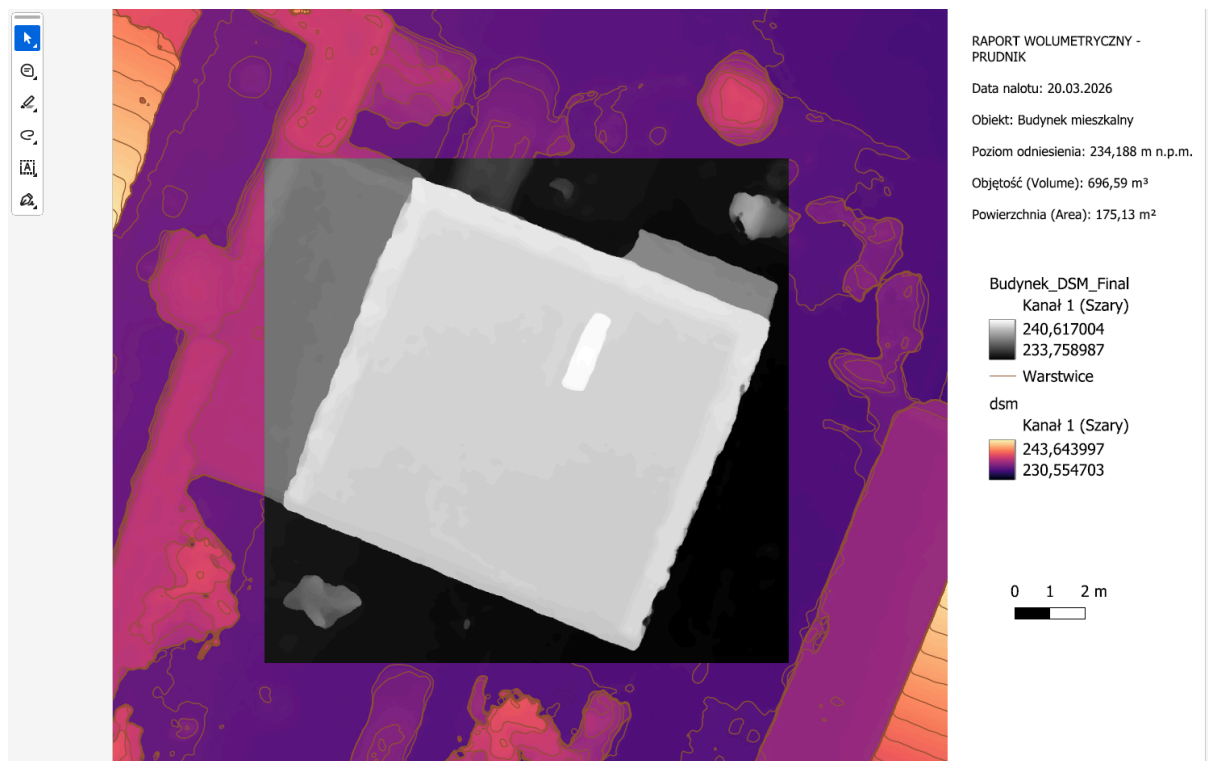
```
6 SUCCESS: Hydrology report automatically exported to:
7 C:/Users/igorh/3D-Objects/Hydrologia/Drainage_Network_Report.csv
8 Ready for Stormwater Management Engineers.
9 -----
```

Project 4: Automated Volumetric Analysis (Earthworks & Stockpiles)

Objective: Provide highly accurate volume calculations for construction earthworks, material stockpiles, and excavation progress reporting.

Method: Utilized a high-resolution Digital Surface Model (DSM) to compute the exact volume of structures and earthworks against a highly precise reference level (e.g., 234.188 m a.s.l.). The workflow automates the extraction of cut/fill metrics directly from drone data.

Result: Verified 3D volumetric extraction delivering actionable engineering data: Volume of 696.59 m³ over a footprint area of 175.13 m².

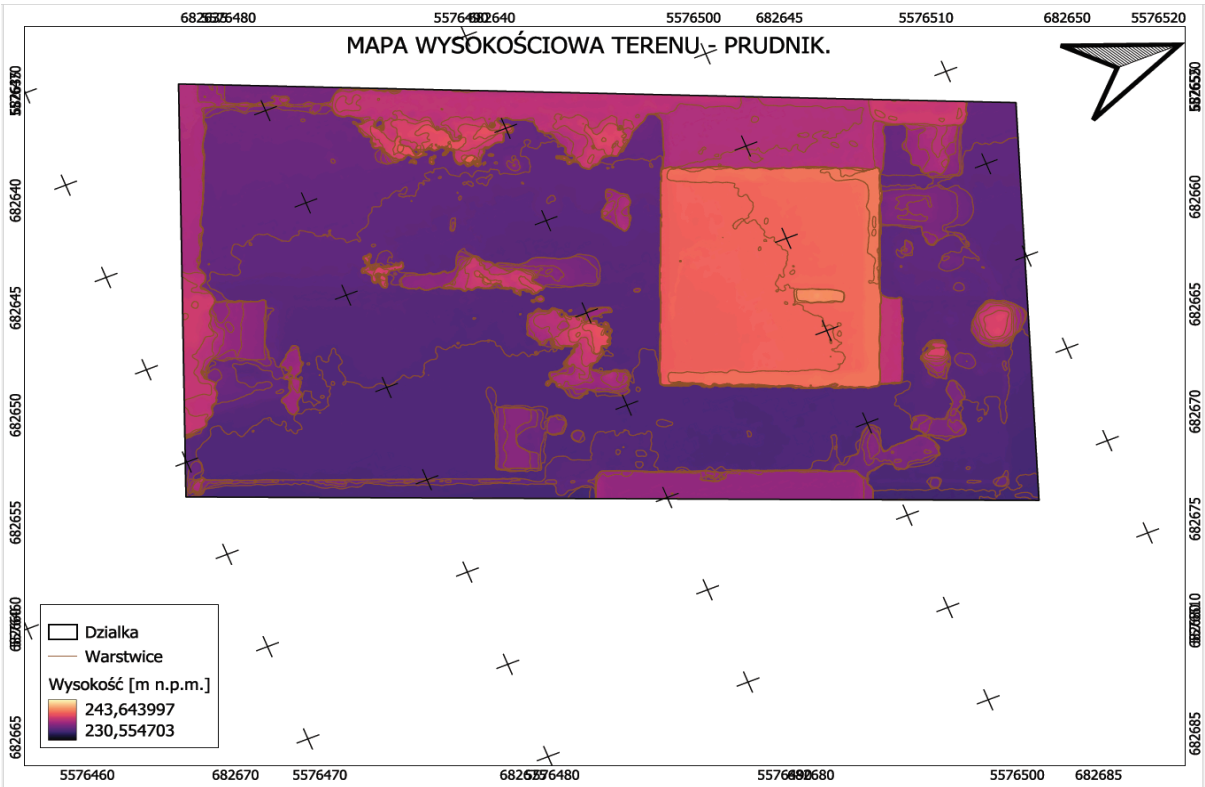


Project 5: Survey-Grade Topographic Mapping & Quality Assurance

Objective: Generate actionable topographic maps and contour data for pre-construction site planning, backed by verifiable spatial accuracy reports.

Method: Processed raw drone photogrammetry data into an accurate Digital Surface Model (DSM), extracting precise elevation contours (ranging from 230.55m to 243.64m). Generated comprehensive Quality Assurance reports detailing tie-points, camera calibration, and absolute GPS errors.

Result: Delivered high-resolution topographic maps with an extremely detailed Average Ground Sampling Distance (GSD) of 1.8 cm, ensuring all deliverables meet engineering tolerances.



Processing Summary

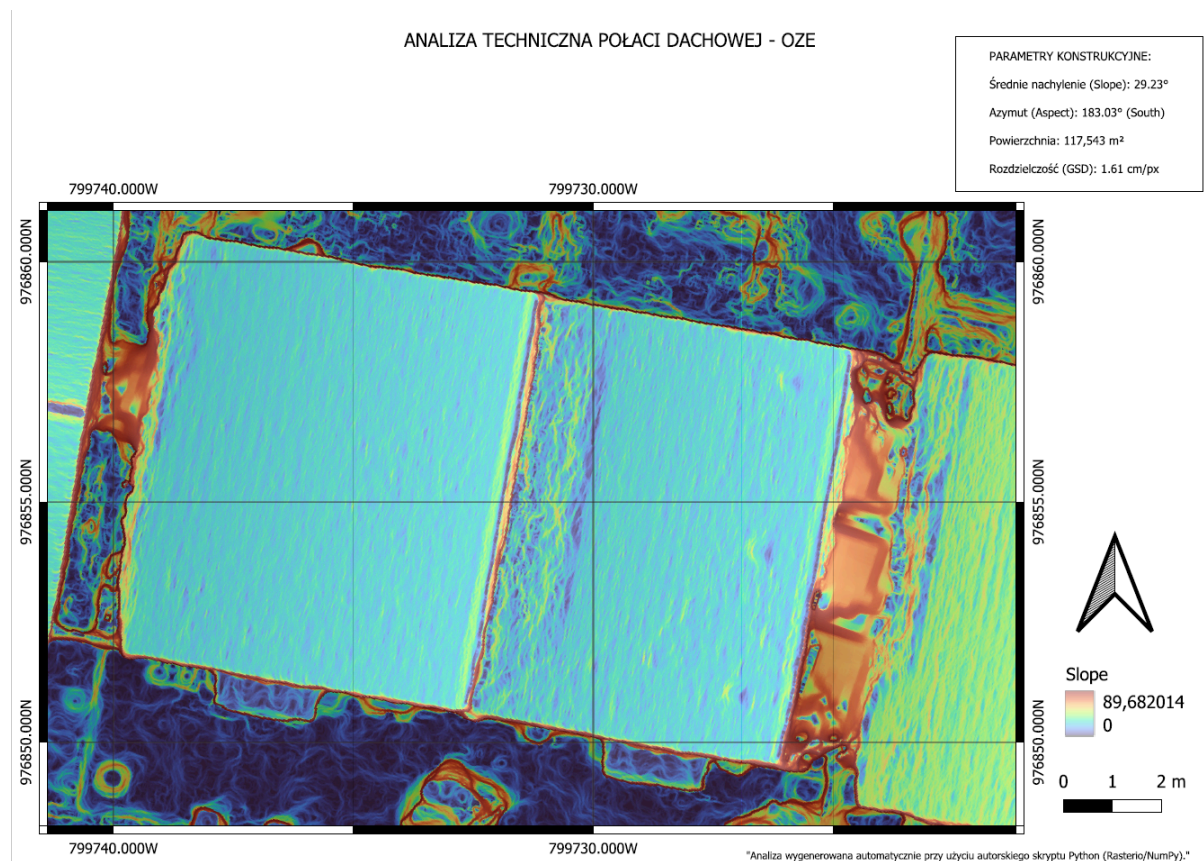
Reconstructed Images	69 over 69 shots (100.0%)
Reconstructed Points (Sparse)	50156 over 59697 points (84.0%)
Reconstructed Points (Dense)	6,900,705 points
Average Ground Sampling Distance	1.8 cm
Detected Features	7,093 features
Reconstructed Features	3,551 features
Geographic Reference	GPS
GPS errors	0.47 meters

Project 6: Automated Solar Potential Analysis (Python / Rasterio)

Objective: Assess rooftop suitability for photovoltaic (PV) installations by automatically extracting critical engineering metrics (tilt and azimuth) from UAV-derived Digital Surface Models (DSM).

Method: Processed high-resolution drone imagery into a precise DSM (GSD 1.61 cm) using WebODM. Developed a custom Python script utilizing Rasterio and NumPy to filter terrain noise and extract exact mathematical averages for slope and aspect from the targeted roof structure. Visualized the technical data in QGIS, producing a survey-grade layout with a georeferenced coordinate grid.

Result: Verified optimal PV conditions: Mean Slope of 29.23° , perfect Southern Aspect of 183.03° , and a usable roof area of 117.54 m^2 . The automated pipeline drastically reduces manual GIS calculation times and delivers client-ready technical reports.



```
--- Analysis for: Fairleigh-Road-1052026-slope.tif ---  
Mean: 29.23  
Max: 89.75  
Std Dev: 21.73  
  
--- Analysis for: Fairleigh-Road-1052026-aspect.tif ---  
Mean: 183.03  
Max: 360.00  
Std Dev: 102.81  
PS C:\Users\igorh\OneDrive\Dokumenty\zdjecia dom UK>
```

Standard Operating Procedures (SOP) & Mission Planning

Objective: Ensure consistent, survey-grade data acquisition by applying rigorous pre-flight planning, hardware QA checks, and optimal photogrammetry parameters.

Methodology: Before any data is processed in WebODM or Python, every flight is meticulously planned and logged. Drawing from my 7-year background in UAV hardware servicing, I enforce a strict checklist system. Key parameters tracked for every mission include:

- **Photogrammetry Overlap:** Maintaining strict front/side overlap (typically 75-80%) to ensure dense point cloud generation without artifacts.
- **GSD Targeting:** Pre-calculating Ground Sample Distance (e.g., 1.29 cm/px at 30m altitude) to meet specific engineering tolerances before take-off.
- **Hardware QA:** Verifying IMU calibration, gimbal angles (Nadir/Oblique), and fixed White Balance for consistent raster processing.

Impact: This systematic approach bridges the gap between field operations and data science, guaranteeing that the input data fed into my Python/GIS pipelines is always reliable, measurable, and free of collection errors.

	A	B	C	D
1	SEKCJA	PARAMETR	WARTOŚĆ	JEDNOSTKA / STATUS
2	METADANE	Data misji	20.03.2026	RR-MM-DD
3		Lokalizacja	Prudnik	
4		Operator		Igor Hajducki
5	1. CHECKLISTA	Baterie (Dron/RC 100%)		TAK
6		Karta SD (Format/V30)		TAK
7		Obiektyw (Czysty)		TAK
8		Strefa (DroneRadar)		TAK
9		Warunki (Wiatr/Opady)		TAK
10	2. PARAMETRY	Wysokość Lotu (H)	30	m
11		Kąt Gimbala	-90	° (Nadir)
12		Interwał (Timed Shot)	2	s
13		Pokrycie Front (Overlap)	75	%
14		Pokrycie Bok (Sidelap)	75	%
15		Ostrość (MF Infinity)		TAK
16		Balans Bieli (Staty)		TAK
17	3. LOGI MISJI	Czas Startu	14:07	GG:MM
18		Czas Lądowania	14:09	GG:MM
19		Czas Misji	2 min	minuty
20		Liczba zdjęć	69	szt.
21		WYLICZONE GSD	1,29	cm/px
22	4. TEREN	Notatki		slonecznie, bez wiatru

	A	B	C	D
1	SEKCJA	PARAMETR	WARTOŚĆ	JEDNOSTKA / STATUS
2	METADANE	Data misji	01.05.2026	RR-MM-DD
3		Lokalizacja	Sunderland	
4		Operator		Igor Hajducki
5	1. CHECKLISTA	Baterie (Dron/RC 100%)		TAK
6		Karta SD (Format/V30)		TAK
7		Obiektyw (Czysty)		TAK
8		Strefa (DroneRadar)		TAK
9		Warunki (Wiatr/Opady)		TAK
10	2. PARAMETRY	Wysokość Lotu (H)	30	m
11		Kąt Gimbala	90 / 45	° (Nadir)
12		Interwał (Timed Shot)	5	s
13		Pokrycie Front (Overlap)	80	%
14		Pokrycie Bok (Sidelap)	70	%
15		Ostrość (MF Infinity)		TAK
16		Balans Bielei (Stały)		TAK
17	3. LOGI MISJI	Czas Startu	14:45	GG:MM
18		Czas Lądowania	14:55	GG:MM
19		Czas Misji	10 min	minuty
20		Liczba zdjęć	31 / 20	szt.
21		WYLICZONE GSD	1,61	cm/px
22	4. TEREN	Notatki		slonecznie, lekki wiatr